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Late Mesozoic and Cenozoic palaeomagnetism of Australia – I. A redetermined apparent polar wander path

Mart Idnurm *Bureau of Mineral Resources, GPO Box 378, Canberra, ACT 2601, Australia*

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Summary. Palaeomagnetic measurements have been carried out on one Cretaceous and five Cenozoic sedimentary sequences of Australia; of those, one is in the Carnarvon Basin of north-western Australia, and the others are in the Otway Basin of south-eastern Australia. The new results are used together with those published previously to define a revised late Mesozoic–Cenozoic apparent polar wander path for Australia. This path differs from earlier, basalt-derived paths by the absence of a large westward excursion and zig-zag irregularities. It is characterized instead by a substantially straight Cenozoic trajectory, a sharp bend in the Late Cretaceous, and a non-uniform rate of apparent polar wander.

The early Tertiary segment of the new path lies north of the original paths, thus eliminating a discrepancy that has been noted previously between the Indian and Australian palaeomagnetic data (Luyendyk & Rennick, Peirce and Klootwyk & Peirce). A re-examination of the original data suggests that this discrepancy, as well as two others in the Australian results, may have been caused principally by incomplete time-averaging of remanence directions, because the volcanics on which the results were based had been extruded episodically.

Key words: Australia, Cenozoic, Cretaceous, DIR, episodic volcanism, poles

1 Introduction

The original Australian apparent polar wander (APW) paths for the late Mesozoic and Cenozoic were determined from measurements on the basalts of eastern Australia (Irving & Green 1958; Robertson 1966; Wellman, McElhinny & McDougall 1969; McElhinny, Embleton & Wellman 1974). The latest and most comprehensive of those paths indicates a net northward movement of the continent through 35° of latitude, and shows a prominent westward excursion from a meridional trend along a zig-zag trajectory. Thus, the general impression from palaeomagnetism is that the drift history for the Australian region during the last 100 Myr was complex. These paths have provided valuable constraints for tectonic reconstructions (e.g. McElhinny 1973; Sclater & Fisher 1974; Luyendyk & Rennick 1977; Norton & Sclater 1979), and for palaeoclimatic and palaeogeographic studies (e.g. Frakes & Kemp 1973; Kemp 1978).

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Concurrently with the palaeomagnetic studies in Australia, Cenozoic palaeomagnetic and seafloor spreading data were being gathered from India and from the eastern Indian Ocean, and by the late 1970s it had become possible to test the self-consistency of the combined Indo–Australian data set. Such comparisons revealed small, but significant, discrepancies. In 1977 Luyendyk & Rennick constructed a hot-spot model for the Ninetyeast Ridge of the eastern Indian Ocean with the aid of palaeolatitudes obtained from the Australian palaeomagnetic poles. The model was in good agreement with Cenozoic geologic data from other aseismic ridges in the Indian Ocean – except within the time interval 32–53 Ma. Luyendyk & Rennick suggested among other possibilities that this discrepancy may be caused by an error in either the age or coordinates of one of the early Tertiary Australian poles that had been derived from the 53 Myr old Barrington volcano. [All K-Ar ages used here have been adjusted to conform with the decay constants of Steiger & Jager (1977).] In 1978 Peirce used the Indian Ocean seafloor spreading data to compare palaeomagnetic directions obtained from the Deccan Traps of India with those from the Barrington volcano, and again found a latitudinal discrepancy, which reached a maximum of 13°. And, in 1979, Klootwijk & Peirce challenged the Barrington volcano results on the basis of additional palaeomagnetic data from India. Klootwijk & Peirce found further that a second Australian pole, obtained from the 34.5 Ma Liverpool volcano, was anomalous as well. Although not noted by them, the position of a third Australian pole, from the 18 Ma Nandewar volcano, also disagrees somewhat with the Indian path (see fig. 3 of Klootwijk & Peirce 1979). More recently, Embleton & McElhinny (1982) have again drawn attention to the discrepancies. The evidence suggested that, for at least the Barrington pole, the discrepancies were due to small errors in the Australian path.

Pending new palaeomagnetic measurements, Klootwijk & Peirce (1979) derived an alternative late Mesozoic–Cenozoic APW path for Australia by using seafloor spreading data to transfer poles from India to Australia. Data transfer between widely different localities is, however, strictly valid only if the geomagnetic field can be adequately represented by a geocentric axial dipole, i.e. without higher-order spherical harmonic terms (e.g. Idnurm 1985). Available evidence suggests that for the Cenozoic era such a model is only approximately valid (e.g. Merrill & McElhinny 1983). Embleton (1981) has used the remanence directions of Australian weathered profiles to define a new, although undated, Cenozoic pole trajectory. These two paths differ from each other in detail, but agree on the absence of zig-zags and any pronounced westward excursion. In each case the authors emphasize that their respective paths are tentative.

It has become increasingly important to eliminate errors in the Australian pole path. First, studies of the behaviour of the time-averaged geomagnetic field rely critically on global coverage of palaeomagnetic data and are therefore hindered by the relative paucity of data from the southern hemisphere (see Merrill & McElhinny 1977). Significant discrepancies in the Australian data would further aggrevate this problem. Secondly, an accurate late Mesozoic–Cenozoic path is required for the use of palaeomagnetism to date lateritic profiles. The existing Cenozoic palaeomagnetic data for Australia have therefore been reassessed, and the APW path has been redetermined from a new series of measurements in combination with earlier results.

2 Discrepancies in the Australian path

Before further measurements it is important to identify the sources of the discrepancies in the Australian basaltic poles. One possibility is the existence of non-dipole components in past geomagnetic fields; this is, however, not supported for the early Cenozoic by the results of geomagnetic studies, nor does it adequately explain the younger discrepancies. Taking Coupland & Van der Voo's (1980) early Tertiary result – that the non-dipole field is essentially quadrupolar and has a Gaussian coefficient ratio $g_2^0/g_1^0 \sim 0.15$ (estimated from their fig. 3) – the

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non-dipole components are found to decrease rather than increase the discrepancy, and only by about 3° (the non-dipole field accentuates mainly the difference in longitudes of the Australian and Indian poles). Similarly, Lee & McElhinny's result (reported in Merrill & McElhinny 1983) of a non-dipole field that has both quadrupole and octupole components fails to account for the discrepancy: those components are of moderate magnitude and their effects tend to cancel. Again for the mid- and late Cenozoic, the non-dipole components do not expain why zig-zag irregularities are present in the Australian pole path, but absent in the Indian path.

As discussed by Embleton & McElhinny (1982), preliminary, unpublished results from detailed remanence studies of Hoffman (see also Hoffman 1981, 1984) and Mangold suggest that the discrepancies in the Barrington volcano results may be due to complex magnetization characteristics of the basalts. Alternatively, the discrepancies may have arisen from an incomplete cancellation of palaeosecular variations when the remanance directions of the individual flows were averaged – despite the large number of flows measured.

In 1970 Wilson pointed out that secular variations are not necessarily cancelled by averaging between successive lava flows, because groups of flows can be extruded over short periods of time. In Wilson's study of Tertiary lavas of Northern Ireland, 32 flows in a sequence of 54 appeared to have been extruded in four brief episodes. Brief periods of basalt extrusion have also been reported from other parts of the world, such as the Stoddart Formation in New Zealand (Evans 1970); Steen Mountains in the USA (Watkins & Baksi 1974); and some Hawaiian lavas (McWilliams, Holcomb & Champion 1982).

Most of the Australian Cenozoic poles had been determined by measurements on one principal extinct volcano, supplemented by fewer measurements on other volcanoes. Thus, four out of six poles on the latest basalt-derived path are based primarily on measurements of the Barrington, Liverpool, Nandewar and Tweed volcanoes. The first three of these, like the lavas reported by Wilson (1970), appear to have successive flows with similar remanence directions. This is most striking for the Nandewar volcano as shown in Fig. 1, where flows with similar directions are

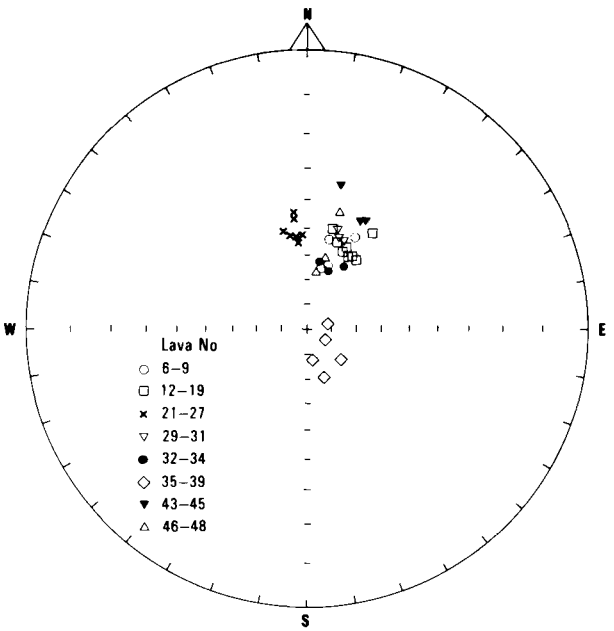


Figure 1. Clustering of directions among successive lavas of the Nandewar volcano. The lavas are numbered up-sequence. All directions are shown in the upper hemisphere. Lavas 46–48 have positive inclinations.

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indicted by the same symbol. To facilitate comparisons, the reversed polarity directions from flows 46–48 have been inverted. Lava flows 35–39 and 43–45 had been excluded from the original analysis because their anomalous directions suggested extrusion during polarity transitions.

As discussed by Wilson (1970), directional grouping could be caused by a rapid extrusion of lavas, or by extrusion at times when the amplitude of secular variations was very small. For the Australian directions the second alternative would imply that the geomagnetic field had behaved discontinuously in the past, shifting abruptly between a number of quasi-static states. Such behaviour contrasts with the continuous oscillatory changes in the geomagnetic field observed in the late Quaternary sedimentary records (e.g. Creer 1974; Creer & Kopper 1976; Turner & Thompson 1979; Barton & McElhinny 1981). Moreover, the second alternative does not explain the grouping of flows 35–39 that deviate by approximately 40° from the overall mean direction for the Nandewar volcano, and were therefore considered to have been extruded during a polarity transition. A recalculation of the Nandewar volcano pole by assignment of unit weights to the directional groups (as previously, flows 35–39 and 43-45 were excluded from the calculation) gives little change from the original pole position, but increases the A_{95} confidence angle from 4.5° to 7.8°. At that level of uncertainty the pole position can no longer be regarded as anomalous.

Although the directional scatter in the Barrington and Liverpool volcanoes is generally much larger than in the Nandewar volcano, 17 out of 34 lava flows used in the Barrington pole calculation and 24 out of 37 Liverpool flows appear to be grouped similarly to those in Fig. 1. Moreover, McElhinny *et al.* (1974) suspected a rapid extrusion of successive lavas in the Nerriga volcanic province where pilot directions from all 12 flows that were thermally demagnetized yielded a single-polarity, tight grouping. They assigned, accordingly, a unit weight to the overall mean direction of the Nerriga flows. Thus, episodicity may have been common in Cenozoic volcanism of eastern Australia.

Irrespective of whether the grouping is caused by episodic volcanism or by quiescence of the geomagnetic field, to eliminate secular variations in the time-averaged field, unit weights need to be assigned to the group means, rather than to individual flow directions. The present series of palaeomagnetic measurements was carried out on sedimentary sequences to avoid any further uncertainties with time-averaging, or with complex magnetization in the eastern Australian lavas.

3 Sampling and measurement procedures

Over 200 samples were collected from six late Mesozoic and Cenozoic sedimentary sequences. Five of the sequences are in the Otway Basin of south-eastern Australia, and the sixth is in the offshore part of the Carnarvon Basin, north-western Australia (Fig. 2). The new measurements are supplemented by palaeomagnetic data from two previous studies in the Otway Basin (Aziz-ur-Rahman 1971; Barton & McElhinny 1981), and by a number of poles from other regions.

The Otway Basin samples were collected as either drillcores or blocks. They were oriented at the outcrop using a magnetic compass and, Sun permitting, also a sun compass. Most of the localities were coastal cliffs where sampling was limited to parts of sequences which were accessible without climbing aids. In the pole calculations unit weights were assigned to mean remanence directions within successive 0.5 m stratigraphic intervals that spanned more than one bed. These intervals are referred to here as *sites* (*cf.* McElhinny 1973). Generally, two or more samples were collected from each site.

The Carnarvon Basin samples were obtained from a petroleum exploration drillcore, North Rankin 1, drilled by Burmah Oil Company Australia Ltd in 1971. The core is ~10 cm in diameter and consists of small, azimuthally-unoriented segments. A hole deviation survey at depth

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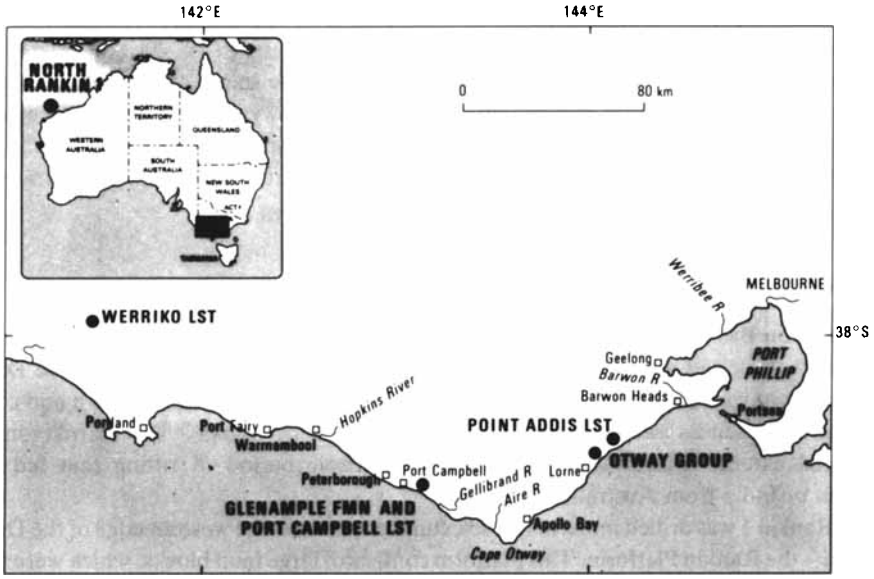


Figure 2. Sample collection localities (filled circles) in south-east and north-west Australia.

intervals of 150 m showed the maximum departure from verticality to be less than 1.3° (BOC of Australia Ltd 1971). Samples were taken from the drillcore at approximately equal intervals of 0.6 m by drilling subcores of 2.5 cm diameter parallel to its axis. The results from each such 0.6 m interval were assigned a unit weight (i.e. given site status).

After preparation, the specimens were sandblasted with fine quartz grit to remove any surface contaminants. The measurements were made on an ScT cryogenic magnetometer. Both alternating field (AF) and thermal demagnetization techniques were used to analyse the remanence, the former in a 2-axis tumbler-demagnetizer described by Senanayake (1981), and the latter in a furnace described by McElhinny, Luck & Edwards (1971). Since rotational remanent magnetization effects (Wilson & Lomax 1972) were often discernible at peaks fields of 20 mT, demagnetization treatments and measurements above 15 mT were repeated by reversing the rotation of the tumbler. All specimens were demagnetized in a minimum of seven steps, and the majority in 10–20 steps. Unless otherwise stated, the remanence directions were obtained from orthogonal projection diagrams by visual fitting of straight line segments.

4 General geology

4.1 OTWAY BASIN

The Otway Basin is a deep, elongated depression filled with Mesozoic and Cenozoic semiconsolidated sediments, sedimentary rocks, and minor volcanics (Wopfner & Douglas 1971; Abele *et al.* 1976). It was initiated during the late Jurassic or Early Cretaceous at the commencement of rifting between Australia and Antarctica. During its early and middle histories the Otway Basin was an intracratonic trough, and accumulated great thicknesses of continental sediments. With the separation of Australia and Antarctica in the Late Cretaceous, the Basin became marginally marine, and at about the same time it was divided by regional block faulting into several sub-basins, each with distinctive sedimentary characteristics. Since the Early Pliocene the Otway Basin has been undergoing uplift; consequently its northern edge is

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now above the sea-level. The uplift has been accompanied by sporadic, but widespread, basaltic volcanism.

The sequences sampled were deposited in the early and late stages of the basinal evolution. Attempts to obtain a pole from mid-evolutionary deposits (Dilwyn Clay and Clifton Formation) were unsuccessful. The previously published results that supplement this study are from the late-stage volcanics (Aziz-ur-Rahman 1971) and from sediments of their associated crater lakes (Barton & McElhinny 1981).

4.2 CARNARVON BASIN

The Carnarvon Basin is located on the western margin of Australia, and comprises a series of sub-basins that extend along 1000 km of the coast. The northernmost of these, the Dampier Sub-basin, from which the present early Tertiary pole was obtained, lies offshore and contains sedimentary sequences ranging from Permian to Holocene in age. The Carnarvon Basin experienced extensive tectonism during a mid-Mesozoic period of rifting that led to the separation of India from Australia–Antarctica.

North Rankin 1 was drilled into a major structural element on the western edge of the Dampier Sub-basin – the Rankin Platform. This platform consists of large fault blocks, which were formed during the mid-Mesozoic extensional phase and have been subsequently draped by sediments. Except for subsidence the Sub-basin appears to have been tectonically quiescent throughout the Tertiary (Powell 1976).

5 Sampling units and results

5.1 OTWAY GROUP

The Otway Group is an Early Cretaceous continental sequence of sandstones, siltstones and minor coal seams. The sediments are predominantly volcaniclastic. The source, or sources, of the volcanic component remain unknown. The minimum age of the Otway Group is constrained by fission track dates on the volcanic detrital components, which range from 126 to 103 Ma (Gleadow & Duddy 1980). The extreme freshness of this detritus, as observed in thin sections, suggests that erosion and deposition had closely followed volcanism.

Twenty-five block and core samples were collected from sandstones within an easterly dipping sequence that outcrops along the Ocean Road at Spout Creek, and at road cuts due east of Spout Creek. The rock is medium-grey and cross-bedded. The top parts of the creek bank and roadcuts are heavily ferruginized: consequently, sampling was restricted to the basal 20 m of section. Localized ferruginization, by seepage, is observed within several laminae lower down the sequence. Although the sampling locality had not been part of the fission track study, its stratigraphic position within the sequence suggests an age of 110–115 Myr (Duddy, private communication).

The NRM intensities ranged from 1.5 to 3.5 mA m^{-1}. Three components of remanence were identified (Fig. 3): a very soft component, which was eliminated at peak fields of 10–15 mT or by heating to 150°C, and is interpreted as recent in origin; an intermediate component of invariably normal polarity; and a hard component of small but variable magnitude and mixed polarity. The small, hard component persists up to at least 590°C. With few exceptions, its direction could not be determined by AF or thermal demagnetization before anharmonic effects or magnetic viscosity made the magnetization unstable. The mixed polarity of the hard component is inconsistent with the long period of Cretaceous normal polarity in which the sediments were deposited, and its high blocking temperature suggests hematite as the carrier of remanence. That

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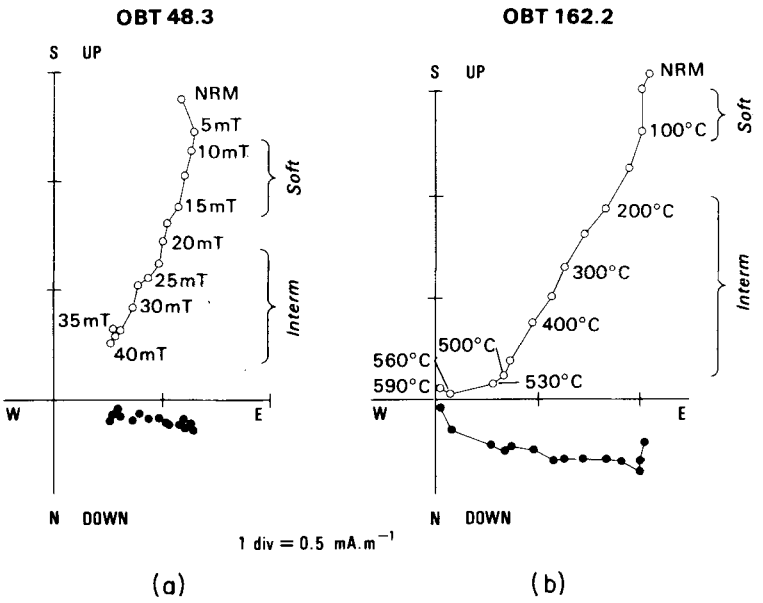


Figure 3. Orthogonal projection plots for two specimens from the Otway Group. Open (solid) symbols show the magnetic vectors projected on to the vertical (horizontal) plane. Tectonic tilt corrections have not been applied.

component is, therefore, probably due to secondary hematite introduced during the ferruginization of the upper parts of the section. The intermediate component is interpreted as the primary remanence. Its normal polarity within the section is consistent with the geomagnetic field during deposition.

Fig. 4 shows the remanence directions (after a correction for a 25° tectonic dip to 95°E) of the intermediate component at 18 sites that yielded well defined directions. A point to note is that, although the cross-beds in the sandstone dip in various directions, the dispersion of the remanence directions in Fig. 4 is small. This implies that depositional inclination errors are negligible, or that the remanence is post-depositional. The mean pole position is: latitude 48.9°S, longitude 148.7°E, A_{95}=3.6° (Table 2).

5.2 POINT ADDIS LIMESTONE

Point Addis Limestone, a greyish-orange to cream calcarenite, was deposited in the eastern-most of the Otway Sub-basins, the Torquay Basin. Planktonic foraminifera indicate a relatively narrow time range, centred on the latest Oligocene (Abele 1979), for this unit.

Fifty-eight samples of the Limestone were collected from coastal cliffs at Point Addis. The samples were from restricted intervals in the top, middle and lower parts of the sequence. Their NRM intensities ranged from 0.07 to 2.3 mA m^{-1}, averaging at 0.2 mA m^{-1}. Twenty-one samples were rejected because demagnetization resulted in a large directional scatter.

Samples from the lower and middle intervals contained a soft, and hence probably recent, component which was generally eliminated by 100°C or 10 mT; an intermediate component; and one or more hard components whose presence was recognized from small intercepts on the orthogonal projection axes. The intermediate component dominates the remanence (Fig. 5b) and is interpreted here as primary.

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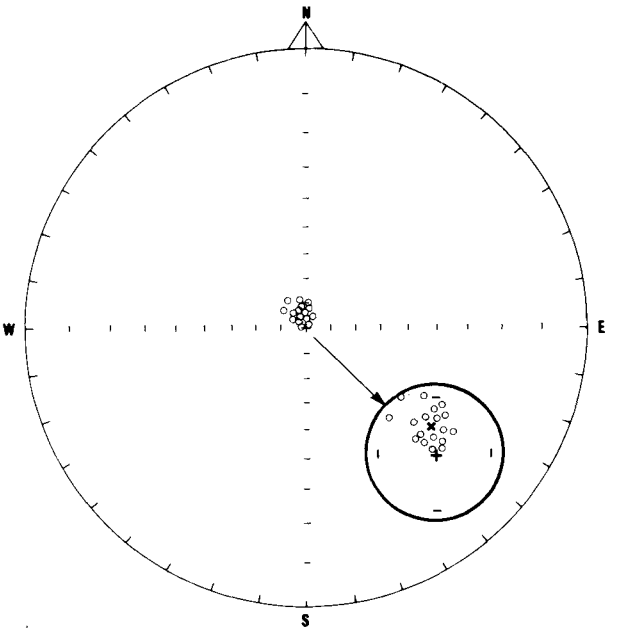


Figure 4. Equal angle stereographic projection of the Otway Group site mean directions for the intermediate component. The directions have been corrected for tectonic tilt. All directions point upwards. Inset shows the central part of the projection expanded. The cross marks the mean direction of the sites.

The uppermost samples were collected at, and below, a prominent sedimentary break (Abele 1979). In contrast to the other samples their characteristic remanence is hard, possibly reflecting a different depositional environment: their remanence showed little or no change in intensity and direction up to the highest demagnetization fields before the directional scatter became excessive (~30 mT), or the highest temperatures before the onset of magnetic viscosity (~350°C). The remanence directions for these samples were therefore determined essentially from vector end points (Fig. 5a). Such estimates may contain small secondary components, as in Fig. 5(b). However, in the absence of evidence for regional tectonism, other than uplift, in the Late Cenozoic, the most likely source for secondary components is iron introduced during weathering or late diagenesis. Such processes are usually prolonged, so that different parts of the rock acquire normal and reversely magnetized components (e.g. Schmidt & Embleton 1976; Idnurm & Senior 1978) and their contribution may, therefore, partly cancel when the sample directions are averaged.

Remanence directions were obtained from 36 samples, representing 21 sites (Fig. 6). Two samples, both from the same site, were rejected because their directions, though well defined, departed by more than four standard deviations from the overall mean direction. The remaining 34 samples yielded a pole of latitude 68.4°S and longitude 118.7°E (A_{95}=4.8°). If the directions from the uppermost sites (seven) are excluded, the mean pole is latitude 68.6°S, longitude 116.0°E, and A_{95}=5.9°, i.e. it does not differ significantly from the mean pole of all 20 sites. From Abele (1979) the mean age of the samples is estimated as appoximately 26 Ma (time-scale of Harland *et al.* 1982). There is no clear field evidence, such as bioturbation, against bedding errors in the remanence. However, Payne & Verosub (1982) find that remanences of sediments like the Point Addis Limestone that contain abundant sand-size grains are post-depositional, and should therefore be free of bedding errors.

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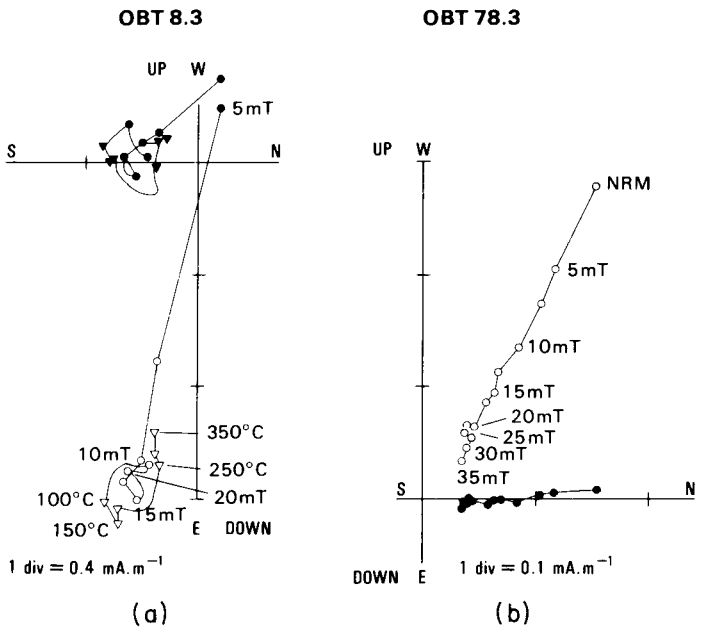


Figure 5. Orthogonal projection plots for two specimens from the Point Addis Limestone. Symbol convention as in Fig. 3.

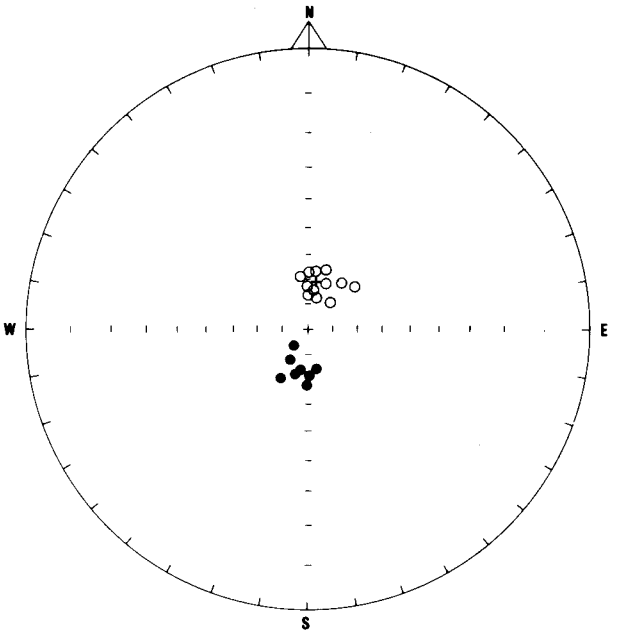


Figure 6. Site mean directions for Point Addis Limestone. Open (solid) symbols denote upward (downward) pointing directions. The cross indicates the mean direction of the sites.

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5.3 PORT CAMPBELL LIMESTONE AND GLENAMPLE FORMATION

These Early and Middle Miocene sequences were sampled east of Port Campbell, where they form sheer coastal cliffs. The sequences represent the latter part of the last major marine transgression in the Port Campbell Embayment of the Otway Basin. The transgression commenced with the deposition of shallow marine clays, which were followed by calcareous muds, and finally, limestones. The transitions between the different sediment types are gradational.

The Glenample Formation, a calcareous clay, represents the middle phase of the transgression. Its thickness in the area sampled is approximately 60 m. The clays are extensively burrowed by marine organisms. The overlying Port Campbell Limestone comprises marls and limestones. Except for some calcarenite beds at the top of the sequence the material is fine-grained. Port Campbell Limestone attains a thickness of approximately 70 m in the sampling region. Detailed studies of planktonic foraminifera by Mallet (1977) at the present sampling localities provide good age controls for both these units.

Thirty-one blue-grey samples were collected from the uppermost part of the Glenample Formation and another 46 blue-grey to pale cream samples were collected from the Port Campbell Limestone. The samples were obtained from five localities along the coast, where the beds are either horizontal or dip gently.

The remanence characteristics of the two units were similar. Their initial intensities of magnetization were low, ranging from 0.02 to 1.1 mA m^{-1} and averaging at 0.2 mA m^{-1}. Thirty samples were rejected because they were either too weak for accurate measurements or gave scattered directions on demagnetization. The remaining 47 samples yielded two components of remanence in addition to a very soft, recent component: these are referred to here as the intermediate and hard components. As in the lower and middle parts of the Point Addis Limestone, the direction of the hard component could not be estimated because the specimens either developed an RRM during AF demagnetization, or became magnetically viscous during heating.

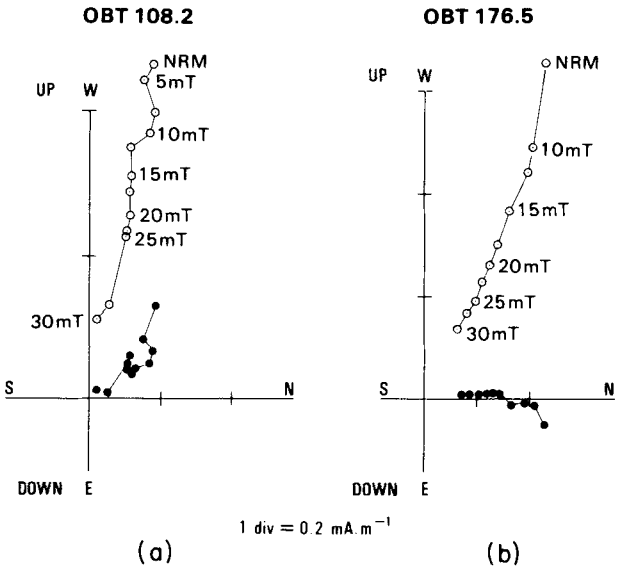


Figure 7. Orthogonal projection plots for two specimens from the Port Campbell Limestone. Symbols as in Fig. 3.

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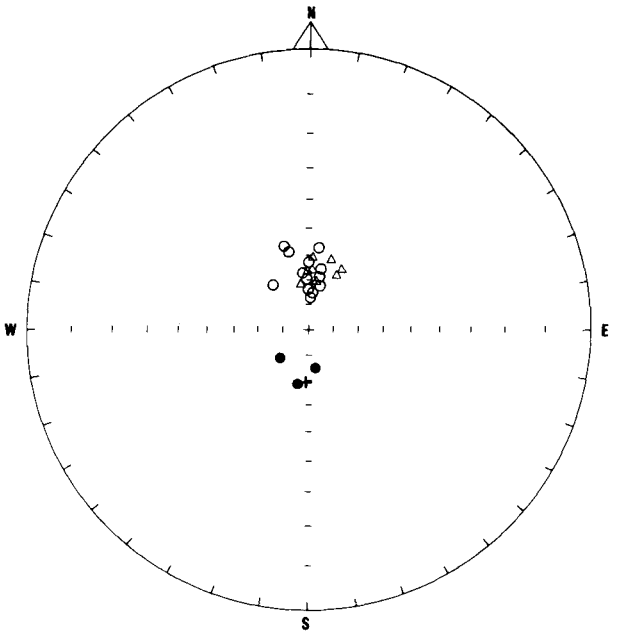


Figure 8. Site mean directions for the intermediate remanence component in Glenample Formation (triangles) and Port Campbell Limestone (circles). Symbols as in Fig. 6.

Typical demagnetization plots are shown in Fig. 7(a, b). Generally the directional scatter between specimens from the same sample was large, angular deviations from the mean being normally 7–10°. The within-sample scatter is reflected in the dispersion of the site mean directions (Fig. 8). The samples which yielded useful results were from three stratigraphic intervals: one within the uppermost Glenample Clay, and the other two within the middle and upper Port Campbell Limestone. The results are listed in Table 1, where the three intervals sampled are indicated in accordance with Mallett's bedding unit nomenclature. The mean ages of the intervals are estimated by interpolating between Mallett's foraminiferal datum levels, defined by *G. druryi* (13.4 Ma), *G. nephenthes* (12.3 Ma); *G. acostaensis* (10.8 Ma); and *G. conomiozea* (tentatively 7.8 Ma).

All, except the topmost sample in C9, and those from C6, have normal polarities. The polarities in the two youngest intervals are consistent with polarity chrons 5 (with possible extension into 4A) and 5r of the reversal column of Harland *et al.* (1982), while the youngest interval would fall within the mixed polarity chrons 5AA and 5AB. In the absence of a more

Table 1. Palaeomagnetic results from Glenample Formation and Port Campbell Limestone.

Bedding Units	Mean Age Estimate (Ma)	Number of Sites	South Pole Lat. (°S)	Long. (°E)	A_{95} (deg)	K
C8 and lower C9	9.5	12	77.2	143.5	6.7	43.175
Upper C6	11	2	67.2	121.4	-	-
Upper G6, C1	13	16	77.4	109.9	5.3	49.744
Combined	12	30	77.2	123.5	4.2	40.703

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complete sequence the correlations must, however, be regarded as very tenuous. The mean age of the 30 sites that yielded reliable directions, and hence the age of the pole, is 12 Ma.

The directional dispersion in the Glenample Formation is moderate, despite extensive bioturbation of the sediment. This suggests that its remanence is post-depositional and should therefore not contain inclination errors due to depositional processes. Bioturbation is not evident in the Port Campbell Limestone. Its mean remanence direction differs, however, from that of Glenample Formation by only 2° and, contrary to expectations from their ages, has the higher inclination. Consequently, if the bioturbated Glenample Formation site directions have no inclination error, the Port Campbell Limestone should also be free of significant error, for otherwise its remanence direction would be shallower than that of Glenample Formation.

5.4 WERRIKO LIMESTONE

This sequence, which comprises mainly flat-lying calcarenites, outcrops along the Glenelg River in the western part of the Otway Basin. It is overlain at some localities by Pliocene basalts. The age of Werriko Limestone has been determined by a combination of foraminiferal and K-Ar basalt dates (Singleton, McDougall & Mallett 1976) as Plio–Pleistocene.

Nineteen samples were collected from a quarry (CRB Quarry of Singleton *et al.* 1976), where approximately 20 m stratigraphic thickness of the sequence is exposed. The outcrop consists of flaggy, orange-brown limestones and interbedded calcareous sands. All except two samples proved unstable under thermal and AF treatments. The two stable samples yielded short, linear segments on orthogonal projection plots, and their remanence directions, therefore, had to be determined essentially from vector end points. Both stable samples were from the basal part of the sequence close to the Plio–Pleistocene boundary. The polarities are reversed in agreement with late Cenozoic reversal stratigraphy.

5.5 NORTH RANKING 1 DRILLCORE

The North Rankin 1 samples were obtained from depths 2276–2293 m below the seabed, and comprise mid-Palaeocene, fine-grained sandstones and sandy siltstones. The rocks are medium-grey with a slight brownish tint, and are extensively bioturbated. Because the Dampier Sub-basin appears to have been tectonically inactive (except for subsidence) during the Tertiary (Powell 1976), tectonic corrections to the remanence directions are assumed negligible. The ambient temperature at 2290 m has been estimated from bottom-hole readings and drill-stem test data as approximately 75°C (BOC of Australia Ltd 1971). The mid-Palaeocene depositional age of the material is obtained from the presence of *G. angulata* and *G. pseudomardi*, which belong respectively to foraminiferal zones P3 and P4 of Blow (1969). The former occurs in the upper part of the sampled core interval, and the latter occurs in the lower part (Wall 1971).

Twenty-seven samples were drilled at approximately equal intervals from 17 m of the core. Two, or occasionally three, specimens were prepared from each sample. The initial remanence intensities ranged from 1.3 to 2.5 mA m^{-1}, and clustered around 1.5 mA m^{-1}. AF demagnetization, supplemented by a number of detailed thermal demagnetization studies, was used to analyse the remanence. Three stable magnetic components besides a viscous component were found in the drillcore (Fig. 9a, b). These are, firstly, a moderately soft component, which was eliminated in peak fields between 20 and 25 mT, or at approximately 100°C. In all except three samples this component has the present-day normal polarity. The three samples with reversed polarities came from well-separated depth intervals, suggesting that either brief episodes of reversed polarity have been recorded in the drillcore, or the downhole directions in these

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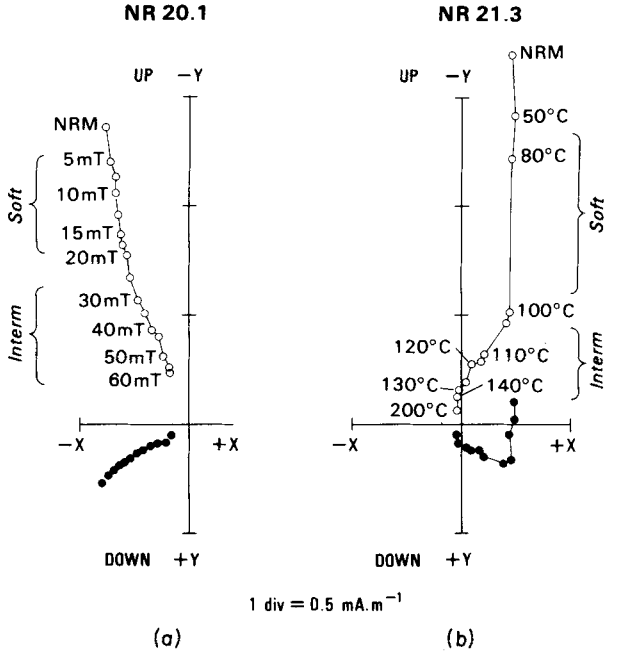


Figure 9. Orthogonal projection plots for two specimens from the North Rankin 1 drillcore. Symbols as in Fig. 3.

three core segments had been marked incorrectly after their recovery. Other evidence, discussed below, strongly favours the second possibility.

The mean angles of inclination of the soft component in 21 samples that gave well defined orthogonal projection segments are shown in Fig. 10(a). The unbiased estimate of the mean

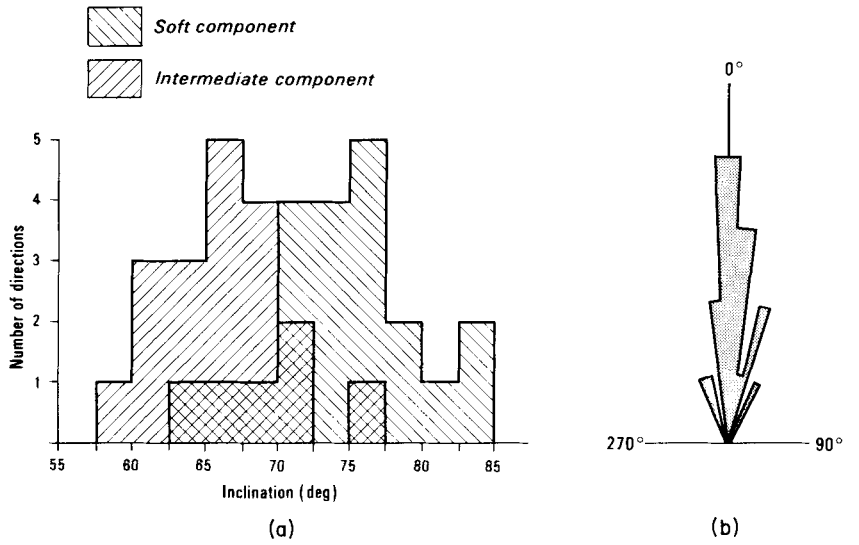


Figure 10. Site mean directions for North Rankin 1 drillcore. (a) Inclination angle histogram for the soft and intermediate hardness components. (b) Rose diagram showing declination differences between the soft and intermediate components in 15 samples.

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inclination, obtained by the iterative method of McFadden & Reid (1982), is 74.6°. In view of the low blocking temperature compared with the estimated *in situ* ambient temperature of 75°C, it seems unlikely that this component is ancient. However, its high inclination angle is inconsistent with recent geomagnetic fields at the drill site. The most likely explanation of its origin is secondary remanence acquired during drilling, being induced by a near-axial magnetic field at the end of the drill stem. The phenomenon of remanence induced by drilling has been studied by Rainbow, Fuller & Schmidt (1972), Burmester (1977), and Lauer (1978), who suggest that in the interior of the drillcore, from where the present samples were obtained, the magnetization is due to viscous remanent magnetization processes (Burmester) or to stress effects produced by mechanical vibrations (Lauer).

The second component is distinguished from the first by a moderate hardness and by a significantly shallower inclination angle (Fig. 10a). Out of 27 samples, two yielded poorly defined orthogonal projection segments for this component, and in five samples the within-sample scatters exceeded 15°. Using again the method of McFadden & Reid, the unbiased estimate for the mean inclination angle of the second component in the remaining 20 samples was obtained as 66.2°, with 95 per cent confidence limits of 63.8° and 68.6°. Because of extensive bioturbation in the sediment, both this and the soft component of remanence must be post-depositional.

If the soft component represents the geomagnetic field at the time of offshore drilling, as suggested above, its declination may be used to orient the individual core segments azimuthally (Van der Voo & Watts 1978; Kodama 1984). Fig. 10(b) shows the declination differences between the first and second components for 15 North Rankin 1 samples from which both components could be extracted. The mean declination difference for these samples is 2.6°, and its standard deviation is 2.9°. Allowing for a regional magnetic deviation of 1°E at the site in 1971, the time of drilling, the mean declination of the second component is 358.4°E, and its 95 per cent confidence limits are 352.5°E and 4.3°E. As noted in Section 8, this estimate is consistent with the locus of poles determined independently from other data. Except in three samples all polarities are normal. The same three samples also gave the discrepant reversed polarities for the first component. This highly unlikely coincidence reinforces the notion that three core segments had been marked incorrectly after drilling, and that the polarities of both the first and second components are constant throughout the sampling interval. The single polarity of the second component suggests that it was acquired relatively rapidly, rather than during any prolonged processes, such as diagenesis. Its magnetic inclination and normal polarity are consistent with magnetization at, or penecontemporaneously with, the time of deposition of the sediments. The mean depositional age of the samples is estimated from the planktonic foraminifera as approximately 58 Ma (time-scale of Harland *et al.* 1982).

The presence of a third remanence component in the North Rankin 1 samples is deduced from small residual intercepts on the orthogonal projection axes. Its origin remains uncertain.

6 Previous pole data

6.1 NEWER VOLCANICS

This suite of basalts was extruded during the Pliocene and Quaternary from numerous and widely distributed vents in central and western Victoria. K-Ar and remanence measurements have been reported previously on many of the volcanics (McDougall, Allsopp & Chamalaun 1966; Aziz-ur-Rahman 1971; Aziz-ur-Rahman & McDougall 1972). The remanences of flows with ages in the range 2–5 Ma were used here to obtain a Pliocene pole (see Appendix). The choice of the relatively narrow age range is considered necessary because the rate of polar wander for the latest Cenozoic turns out to be rapid (Fig. 12). Bearing in mind the

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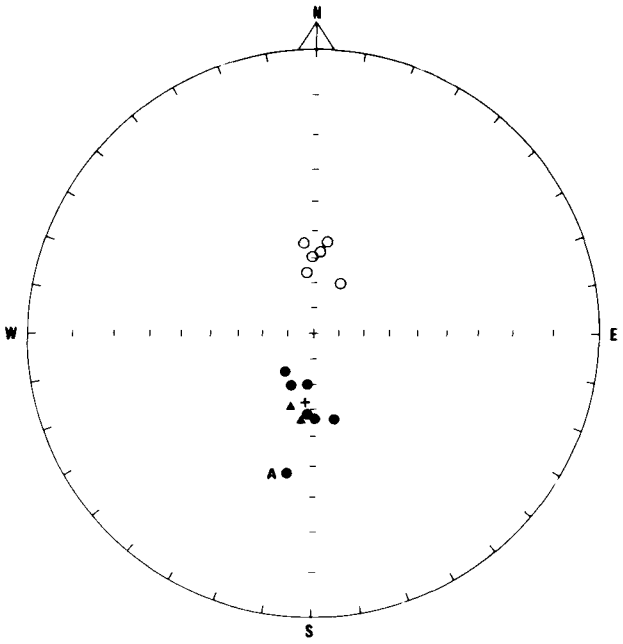


Figure 11. Site mean directions for the Newer Volcanics (circles) and Werriko Limestone (triangles). The direction labelled by A was omitted from the analysis. Symbols as in Fig. 6.

possibility that the volcanism may have been episodic, unit weights were assigned to the mean direction of (a) each volcano, and (b) different volcanoes with similar K-Ar ages. This yields the directions shown in Fig. 11, which also includes the two Werriko Limestone directions from the present study. It should be noted that the Newer Volcanics and the Werriko Limestone have similar directions. Consequently, if inclination errors in the volcanics may be assumed negligible, then any inclination errors in the Limestone should also be small.

With one exception the directions in Fig. 11 form a close group. The single scattered direction lies more than three circular standard deviations from the centre of the group and is therefore regarded as anomalous. It may, for example, represent a geomagnetic reversal transition. The remaining 14 directions give a pole at 83.5°S, 103.6°E (Table 2). Assuming that the age of the Werriko Limestone samples is 2.0 Ma (Plio–Pleistocene boundary date), the mean age of the sites is 2.9 Ma.

6.2 HOLOCENE LAKE SEQUENCES

A Holocene pole was obtained from the secular variation records of Barton & McElhinny (1981) on sediment cores from three volcanic crater lakes in south-western Victoria: Lakes Keilambete, Gnotuk and Bullenmerri. Age determinations were based on sequences of radiocarbon dates that span the last 10000 yr. Assignment (perhaps somewhat conservatively) of unit weights to the mean remanence directions of successive 500 yr intervals in each lake yields the Holocene pole 89.4°S and 144.6°E (A_{95}=1.5°). Long-term laboratory redeposition of sediments from Lakes Bullenmerri and Keilambete yielded no inclination errors (Barton & McElhinny 1979).

7 The redetermined pole path

The new path for late Mesozoic and Cenozoic Australia is shown in Fig. 12. Table 2 summarizes the pole data. The path includes several poles not discussed in the foregoing: a 100 Ma pole from

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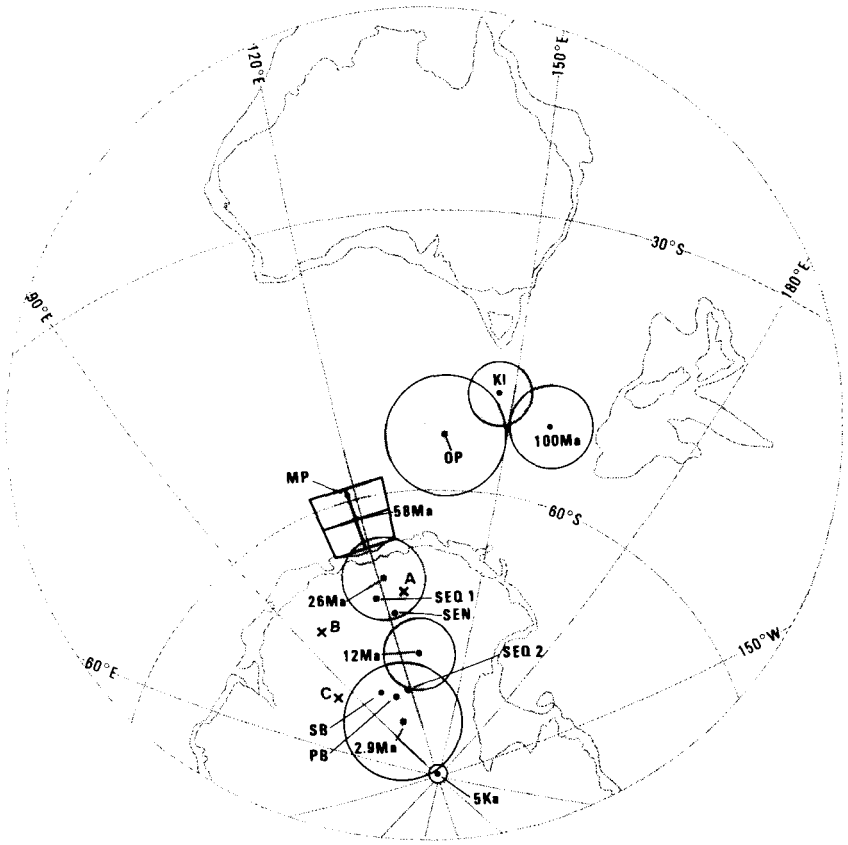


Figure 12. Redetermined late Mesozoic–Cenozoic APW path for Australia. The new poles are labelled 5 Ka, 2.9 Ma, 12 Ma, 26 Ma, 58 Ma and Kl. Points SB, PB, MP, SEQ1, SEQ2, and SEN denote undated poles from lateritic (weathered) profiles. Pole OP represents a thermal overprint that has affected the southeastern margin of Australia. The 100 Ma pole is from Embleton (1981). Crosses marked A, B and C show respectively the pole positions from the Barrington, Liverpool and Nandewar volcanoes which are now considered anomalous. Ages alongside the poles are the estimated mean ages of sites. The age of the Otway Group pole, Kl, is estimated to lie in the range 110–115 Ma. Equal angle projection.

Embleton (1981); the mean, OP, of five overprint poles from south-eastern Australia (Schmidt & Embleton 1981); and six poles from ferruginous weathered profiles of various parts of Australia. The poles from the Barrington, Liverpool and Nandewar volcanoes (McElhinny *et al.* 1974) are shown for comparison.

The overprint magnetizations that give pole OP have been attributed by Schmidt & Embleton (1981) to a thermal event which affected the south-eastern margin of Australia and marked the initiation of seafloor spreading in the Tasman Sea. The occurrence of this event is supported by fission track measurements on apatites in Palaeozoic granites: the track ages have been reset to 80–126 Ma, and their lengths indicate that the rock temperatures were 100–125°C at 70–80 Ma (Moore, Gleadow & Lovering 1983). Thus, the minimum age of the pole OP is 70 Ma. On the other hand, its maximum age must be less than 100 Ma, the age of the Late Cretaceous pole which flanks it in the east.

The weathered profiles are undated: consequently, their remanence directions help to define the APW trajectory, but not the rate of polar wander. Each pole is based on more than 30

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Table 2. Summary of new and recalculated palaeomagnetic results.

Rock Unit	Site Coordinates		No. of Sites (Samples)	Estimated Mean Age (Ma)	Remanence Direction				South Pole			
	Lat. (°S)	Long. (°E)			Decl. (deg.)	Incl. (deg.)	k	α_{95} (deg.)	Lat. (°S)	Long. (°E)	K	A_{95} (deg.)
Otway Gp.	38.4	144.2	18(25)	K1	344.5	-84.2	359.9	1.8	48.9	148.7	97.3	3.6
North Rankin 1	19.6	116.1	20(20)	58	358.4	-66.2	-	**	61.7	118.4	-	***
Point Addis Lst.	38.4	144.3	20(34)	26	16.5	-71.8	130.6	2.9	68.4	118.7	47.8	4.8
Glenample Fm./ Port Campbell Lst.	38.7	143.1	30(48)	12	6.1	-67.3	89.7	2.8	77.2	123.5	40.7	4.2
Newer Volcanics/ Werriko Lst.	38.0	142.5*	14(58)	2.9	5.4	-61.2	84.2	4.4	83.2	103.6	42.5	6.2
Holocene lake sediments	38.2	143.0*	57(-)	0.005	0.0	-57.9	229.7	1.3	89.4	144.6	168.5	1.5

* Mean coordinates of sites
** $dp = 4.8°$, $dm = 5.9°$
*** $dp = 3.3°$, $dm = 8.0°$

independent directions. Only poles PB, for the Perth Basin (Schmidt & Embleton 1976); SB, for the Springfield Basin (Schmidt, Currey & Ollier 1977); and MP, for the Morney Profile (Idnurm & Senior 1978), have been previously published. The others will be reported in conjunction with separate studies. To avoid congestion, the 95 per cent confidence circles for the laterite profile poles have been omitted in Fig. 12. The diameters of all those circles are less than 4°.

8 Discussion

It should be noted, first, that the longitude of the 58 Ma pole, which was estimated on the assumption that the soft remanence component in the North Rankin 1 samples represents the present field, is consistent with the early Tertiary trajectory as defined by the other poles, thus supporting that assumption. Secondly, the latitude of the 58 Ma pole is displaced considerably northward of the 53 Ma pole from the Barrington volcano, in agreement with the results from India and from the eastern Indian Ocean.

Within the limits of uncertainty of the pole estimates, the Cenozoic path for Australia is straight and meridional, except possibly for a slight westward excursion in the latest Cenozoic. The trajectory appears to be unaffected, for example, by the collision of India and Asia in the late Palaeocene or early to middle Eocene (Klootwijk & Peirce 1979; Patriat & Achache 1984), by the change of course of the Pacific plate approximately 43 Ma (G. B. Dalrymple, in Butler & Coney 1981), or by the change in the rate of seafloor spreading between Australia and Antarctica during the middle Eocene (Cande & Mutter 1982). The trajectory has, instead, a sharp bend in the Late Cretaceous. The trajectory and the timing of its bend are discussed elsewhere (Idnurm, in preparation) in terms of non-dipole effects and true polar wander.

Another prominent feature of the new path is a marked variation in the rate of pole moment during the Cenozoic Era (Fig. 12). The variable rate, and its geomagnetic and geodynamic implications, are discussed in Part II (Idnurm 1985).

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for reconstructing continents, and A. Murray adapted the program to run on CYBER. Part of the magnetometer data was processed with the aid of computer programs developed by J. W. Giddings. J. Stelzer assisted with the sampling, which encompassed some unfavourable physical environments. M. Steele drew the diagrams. The Victorian National Parks Service gave permission to sample in the Port Campbell and Otway National Parks, and provided staff to advise on access to sampling localities. The drillcore from North Rankin 1 was made available for measurements by the Core and Cuttings Laboratory, Bureau of Mineral Resources. This paper is published with the permission of the Director, Bureau of Mineral Resources.

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Appendix: Pliocene pole data

The Pliocene pole was calculated from the mean directions of 18 Newer Volcanics localities, supplemented by two directions from Werriko Limestone. For the Newer Volcanics the following sites of Table 1, Aziz-ur-Rahman (1971), were used: 43; (5, 17, 48); (20, 21, 22, 25); 30; 31; (35, 36, 37); 27; 33; 32; 39; 16. The parentheses indicate localities where directions have been averaged for reasons discussed in the text. The two Werriko Limestone site directions are: (1) declination 187.0°, inclination 55.7°, and (2) declination 196.5°, inclination 59.5°.

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